

D602

Deployment

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

Stakeholders at a 250-clinic dental network list requirements for deploying a no-show prediction model: it must flag patients likely to miss appointments, return scores within 1 second, and be available during clinic hours. The network's logo was updated recently. Which is a functional requirement?

- A. Match the updated logo on the scoring screen.
- B. Flag patients likely to miss appointments.
- C. Be available during clinic hours.
- D. Return scores within 1 second.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A retailer with 1,100 stores needs product recommendations refreshed once each night for the next day's marketing emails. The data science team uses Python 3.12. Which deployment pattern fits?

- A. A real-time API that scores each request in under 50 ms
- B. A continuous streaming pipeline
- C. Scheduled nightly batch scoring
- D. Upgrade the Python version first

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A card issuer must approve or decline each purchase within 200 milliseconds at checkout. Which deployment pattern fits?

- A. A weekly fraud report
- B. Nightly batch scoring
- C. Batch, because batch is always cheaper
- D. A real-time (online) inference API

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A model that ran fine on a data scientist's laptop fails in production because the server has a different version of a numeric library. The team has 9 members across two offices. Which practice best prevents this?

- A. Upgrade every laptop to the newest library.
- B. Package the model in a container with pinned dependency versions.
- C. Move the team into one office.
- D. Retrain the model on the server.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A fraud-scoring API at a payments firm with 14 million cardholders has an SLA of 99.9% availability over a 30-day month. The API runs in two cloud regions. What is the maximum downtime allowed per month?

- A. About 43.2 minutes
- B. About 4.3 minutes
- C. About 86.4 minutes
- D. About 8.8 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A service was down for 7.2 hours in a 30-day month (720 hours). What was its availability?

- A. 1.0%
- B. 92.8%
- C. 99.9%
- D. 99.0%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A pricing API at a travel site is stateless, and its traffic spikes 10 times during holiday sales. The site's mascot is a cartoon owl. The team wants to handle the spikes and avoid a single point of failure. Which approach fits?

- A. Scale down during sales to control cost
- B. Move to one much larger server
- C. Horizontal autoscaling of instances behind a load balancer
- D. Throttle all users permanently

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A team replaced a 16-core database server with a 64-core server. Which kind of scaling is this?

- A. Autoscaling
- B. Horizontal scaling
- C. Vertical scaling
- D. Load balancing

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A deployment script for a claims model, kept in a shared Git repository with 40 contributors, contains a hard-coded database password. The script runs every 15 minutes. What is the best fix?

- A. Make the repository private.
- B. Rotate the password, move it to a secrets manager read at runtime, and use a least-privilege account.
- C. Run the script less often.
- D. Encrypt the server's disk.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A demand model at a beverage company, deployed a year ago with 8% forecast error, now shows 19% error after consumers shifted toward low-sugar drinks. The model runs on a Kubernetes cluster with 6 nodes. What is happening, and what is the response?

- A. Concept drift; retrain on recent data and keep monitoring.
- B. Add nodes to the cluster.
- C. Nothing; a validated model never needs retraining.
- D. Redeploy the same model, since it is probably a server bug.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Monitoring shows the distribution of customer ages feeding a model has shifted, but the relationship between inputs and the outcome has not changed yet. What is this called?

- A. No issue, since drift cannot matter while accuracy holds
- B. Concept drift
- C. Overfitting
- D. Data drift

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A bank wants to release a new version of its credit model to 3 million users while limiting risk. The release is planned for a Tuesday. Which strategy fits?

- A. A canary release: route a small share of traffic, monitor, then expand or roll back.
- B. Move the release to a Friday.
- C. Replace the old model for all users at once.
- D. Test only in development, since production testing is never acceptable.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

Stakeholders at a 250-clinic dental network list requirements for deploying a no-show prediction model: it must flag patients likely to miss appointments, return scores within 1 second, and be available during clinic hours. ~~The network's logo was updated recently.~~ Which is a functional requirement?

Target word: the functional requirement **Decides:** What the system does vs. how well.

A	Camouflage	Not a stated requirement.
B	Correct	Describes the system's behavior.
C	Adjacent category	Availability is non-functional.
D	Confusion set	Latency is a quality attribute, non-functional.

2. Answer C Concept 2

A retailer ~~with 1,100 stores~~ needs product recommendations refreshed once each night for the next day's marketing emails. ~~The data science team uses Python 3.12.~~ Which deployment pattern fits?

Target word: inference pattern **Decides:** Results are needed once per night.

A	Adjacent category	Low latency is not needed for next-day emails.
B	Confusion set	For event-by-event needs, not a nightly refresh.
C	Correct	Matches the once-a-day need at lowest cost and complexity.
D	Camouflage	The language version does not decide the pattern.

3. Answer D Concept 2 · transfer

A card issuer must approve or decline each purchase within 200 milliseconds at checkout. Which deployment pattern fits?

Target word: inference pattern — the reverse case **Decides:** A decision per request in milliseconds.

A	Adjacent category	Reporting, not a decision at checkout.
B	Inverted operation	Results arrive a day late.
C	Absolute language	Cost does not override the latency requirement.
D	Correct	Only online scoring can answer during the transaction.

4. Answer B Concept 3

A model that ran fine on a data scientist's laptop fails in production because the server has a different version of a numeric library. ~~The team has 9 members across two offices.~~ Which practice best prevents this?

Target word: prevent environment mismatch **Decides:** Dependency versions differ between environments.

A	Inverted operation	Moves the mismatch instead of pinning versions.
B	Correct	The same environment runs everywhere.
C	Camouflage	Location does not control library versions.
D	Adjacent category	Does not fix the library mismatch for the next release.

5. Answer A Concept 4

A fraud-scoring API at a payments firm with 14 million cardholders has an SLA of 99.9% availability over a 30-day month. ~~The API runs in two cloud regions.~~ What is the maximum downtime allowed per month?

Target word: allowed downtime **Decides:** 0.1% of 43,200 minutes.

A	Correct	$30 \times 24 \times 60 = 43,200$ minutes $\times 0.001 = 43.2$.
B	Confusion set	That is 99.99% (four nines).
C	Camouflage	Doubles the budget for two regions; the SLA is for the service.
D	Adjacent category	That is 99.9% over a full year.

6. Answer D Concept 4 • transfer

A service was down for 7.2 hours in a 30-day month (720 hours). What was its availability?

Target word: solve for availability **Decides:** Uptime 712.8 of 720 hours.

A	Inverted operation	That is the downtime share.
B	Wrong stem number	Subtracts 7.2 from 100 as if hours were percent.
C	Confusion set	99.9% would allow only about 0.72 hours.
D	Correct	$712.8 / 720 = 0.99$.

7. Answer C Concept 5

A pricing API at a travel site is stateless, and its traffic spikes 10 times during holiday sales. ~~The site's mascot is a cartoon owl.~~ The team wants to handle the spikes and avoid a single point of failure. Which approach fits?

Target word: scaling approach **Decides:** Stateless service, spiky load, no single point of failure.

A	Inverted operation	Removes capacity when it is most needed.
B	Adjacent category	Vertical scaling keeps a single point of failure.
C	Correct	Adds instances with demand and removes the single server as a failure point.
D	Absolute language	Blocks legitimate demand year-round.

8. Answer C Concept 5 • transfer

A team replaced a 16-core database server with a 64-core server. Which kind of scaling is this?

Target word: name the scaling type **Decides:** A bigger single machine.

A	Adjacent category	Autoscaling adjusts capacity automatically; this was a manual swap.
B	Inverted operation	Horizontal adds more machines.
C	Correct	Adding power to one machine is vertical.
D	Confusion set	Spreads traffic across machines; there is still only one.

9. Answer B Concept 6

A deployment script for a claims model, kept in a shared Git repository with 40 contributors, contains a hard-coded database password. ~~The script runs every 15 minutes.~~ What is the best fix?

Target word: secure credential handling **Decides:** A secret is exposed in code and history.

A	Dropped element	The secret remains in history and is never rotated.
B	Correct	Removes the exposure, invalidates the leaked value, and limits damage.
C	Camouflage	Frequency does not affect the exposure.
D	Adjacent category	Protects stored data, not a password shared in code.

10. Answer A Concept 7

A demand model at a beverage company, deployed a year ago with 8% forecast error, now shows 19% error after consumers shifted toward low-sugar drinks. ~~The model runs on a Kubernetes cluster with 6 nodes.~~ What is happening, and what is the response?

Target word: diagnose decay and respond **Decides:** Customer behavior changed how inputs relate to demand.

A	Correct	The relationship the model learned no longer holds.
B	Camouflage	Compute capacity does not affect accuracy.
C	Absolute language	Models decay as the world changes.
D	Dropped element	Ignores the behavior change.

11. Answer D Concept 7 · transfer

Monitoring shows the distribution of customer ages feeding a model has shifted, but the relationship between inputs and the outcome has not changed yet. What is this called?

Target word: name the drift type **Decides:** Inputs shifted; the relationship did not.

A	Absolute language	Data drift is an early warning worth tracking.
B	Confusion set	That is a change in the input-outcome relationship.
C	Adjacent category	A training problem, not a production shift.
D	Correct	A change in input distributions.

12. Answer A Concept 8

A bank wants to release a new version of its credit model to 3 million users while limiting risk. ~~The release is planned for a Tuesday.~~ Which strategy fits?

Target word: release strategy **Decides:** Large user base and a need to limit risk.

A	Correct	Limits exposure and allows quick rollback.
B	Camouflage	The day of the week is not the risk control.
C	Inverted operation	Maximizes exposure if something is wrong.
D	Absolute language	Controlled production exposure is standard practice.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Functional vs. non-functional deployment requirements	Functional requirements say what the deployed model does; non-functional requirements set quality levels such as latency, availability, security, and scale.	Treating response time or uptime as a feature.	“Within 1 second,” “available during,” “encrypted.”
2	Batch vs. real-time inference	Use scheduled batch scoring when results are needed on a cycle; use an online API when each request needs an answer in milliseconds.	Building a low-latency API for nightly needs, or batch scoring for decisions made at checkout.	How soon after the event the prediction is needed.
3	Reproducible environments	Package code with pinned dependency versions (for example in a container) and version code, data, and models so production matches development.	Fixing “works on my laptop” problems by retraining or upgrading machines.	Same code behaving differently on another machine.
4	Availability math	Allowed downtime = $(1 - \text{availability}) \times \text{period}$; 99.9% over 30 days $\approx$ 43.2 minutes.	Miscounting the nines or using a yearly figure for a monthly SLA.	An SLA percentage and a time period.
5	Horizontal vs. vertical scaling	Horizontal scaling adds instances behind a load balancer (suits stateless services and removes single points of failure); vertical scaling makes one machine bigger.	Calling a bigger server “horizontal,” or relying on one large server for spiky load.	More machines vs. a bigger machine.
6	Deployment security basics	Keep secrets out of code (use a secrets manager), rotate any exposed credential, grant least privilege, and encrypt in transit and at rest.	Thinking a private repo or disk encryption fixes a hard-coded password.	Credentials visible in scripts or config files.
7	Monitoring for drift	Data drift = input distributions shift; concept drift = the relationship between inputs and target changes; both call for monitoring and retraining.	Assuming a validated model stays accurate, or confusing the two drift types.	Performance decay after a change in behavior or population.
8	Low-risk release strategies	Canary or blue-green releases expose a new version to limited traffic, monitor it, and allow quick rollback.	Switching all traffic at once, or testing only outside production.	A new model version for a large user base with a need to limit risk.

Course scope (public description): Deployment is the practice of operationalizing data analysis within a business environment. Given an analysis, learners determine the business functional and non-functional requirements for wider use and implement pipelines and functions to deploy analyses at scale. This course discusses topics such as security, scalability, usability, and availability. The following courses are prerequisites: Analytics Programming, Data Management, Data Preparation, and Statistical Data Mining.